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A review on the integration of radiative cooling and solar energy harvesting

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ABSTRACT

With developments in material science and thermal engineering, radiative sky cooling has received renewed interest as it provides passive cooling energy during both day and night. Similarly, solar energy harvesting systems, such as photovoltaics, photovoltaic/thermal, and thermal collectors, have been used extensively to meet global energy demands. Integrating the individual systems into the combined solar energy–radiative cooling (SE–RC) system provides multifunctionality, increased operating time, and higher energy gains per unit area. From initial studies to the latest developments, this review has analyzed such integrated systems considering their module configurations, design techniques, applied materials, operating conditions, and performance results. Despite their impressive potential, these systems encounter challenges of low cooling power, stringent spectral requirements, and varying energy supply and demand. Recent progress made to address such challenges, potential solutions, and future research directions have also been discussed. Considering the global energy crisis and environmental issues, the cooling, heating, and electricity generated by SE–RC systems will be used in diverse applications like buildings and vehicles. Therefore, it is anticipated that this review will serve as a guide for integrating solar energy and radiative cooling, and will help identify patterns that are not evident from single studies so to further promote research in this domain.

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1. Introduction

With the improvements in human living standards along with the steep rise in population, there has been an ever-increasing demand for energy consumption. Particularly, energy demand for cooling applications has been growing at an alarming rate, and it is expected that by 2050, the demand in commercial and residential sectors may rise by about 260% and 780%, respectively [1]. At present, the widely used vapor compression-based cooling systems adversely affect the environment (i.e., global warming) and consume huge resources (i.e., electricity input) [2]. These concerns have motivated researchers to look for alternate cooling techniques, which are environmentally friendly and energy-efficient.

Outer space has great potential to be used as a heat sink as it has a very low temperature (~3 K). According to the Stefan–Boltzmann law, radiation is emitted from all bodies at temperatures above absolute zero, and the higher a body's temperature, the more radiation would be emitted in a particular wavelength band [3]. As the earth's surface and most terrestrial objects have a temperature of around 300 K, thermal radiation from the earth is mostly emitted in the mid-infrared (MIR) wavelength band (4–25 μm) [4]. Owing to the large temperature difference, the earth's heat is transferred into space through thermal radiation, eventually keeping the earth's temperature to remain habitable. Water vapor and atmospheric gases reduce this heat transfer; however, in the wavelength band from 8 to 13 μm (i.e., atmospheric window [AW]), the atmosphere is highly transparent and allows thermal radiation to pass through (Fig. 1a) [5].

As the AW coincides with the MIR band, this spectral range can be used by sky-facing terrestrial objects to radiate their heat to outer space and eventually cool themselves [4,6–8]. The passive cooling technique of radiating heat to the sky is known as radiative cooling (RC) and it is both environmentally friendly and energy-efficient. In contrast to the conventional cooling techniques, it does not dump the waste heat into the atmospheric air or local water bodies, and also does not take require any input energy [9].

RC has been widely studied because of its potential for passive thermal management in various applications (e.g., buildings,

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Nomenclature

Abbreviations

a/e	absorber/emitter
ARC	antireflection coating
AW	atmospheric window
c-Si	crystalline silicon
CSP	concentrated solar power
MIR	mid-infrared
PDMS	polydimethylsiloxane
PE	polyethylene
PV	photovoltaic
PVT	photovoltaic thermal
PWV	precipitable water vapor
RC	radiative cooling
RCer	radiative cooler
RH	relative humidity
SE	solar energy
SiO_2	silicon dioxide/silica
SRC	selective reflection cooling
TPV	thermophotovoltaic
TWC	total water column

Symbols

β	PV temperature coefficient
K	thermal conductivity
ΔT	temperature difference
$\dot{m}$	mass flow rate
G_{sun}	solar radiation
A_{RCer}	area of radiative cooler
h_{conv}	convective heat transfer coefficient
$h_{conv,btm}$	heat transfer coefficient on bottom
$h_{conv,top}$	heat transfer coefficient on top side
I_{sc}	short-circuit current
L_{air}	air gap thickness
L_{ins}	insulation thickness
T_{amb}	ambient temperature
T_{cell}	solar cell temperature
T_{in}	fluid inlet temperature
T_{out}	fluid outlet temperature
T_{plate}	absorber/emitter plate temperature
T_{PV}	PV temperature
T_{ref}	reference temperature
T_s	RC surface temperature
V_{oc}	open-circuit voltage
v_{wind}	wind speed
α_{plate}	absorptivity of absorber/emitter plate
ε_{atm}	emissivity of atmosphere
ε_{plate}	emissivity of absorber/emitter plate
ε_{RCer}	emissivity of radiative cooler
Θ_{azi}	azimuthal angle
Θ_{inc}	incident angle with surface normal
Θ_{sur}	surface inclination angle with ground
τ	transmissivity
η_{ref}	PV reference efficiency
c_p	specific heat capacity
G_{sun}	total solar radiation
P_{atm}	atmospheric radiation power
P_{elec}	electrical power generated
$P_{net,rad}$	net radiative cooling power
$P_{non,rad}$	non-radiative heat transfer power
P_{rad}	thermal radiation power emitted
P_{sun}	solar radiation power absorbed
P_{ther}	thermal power collected

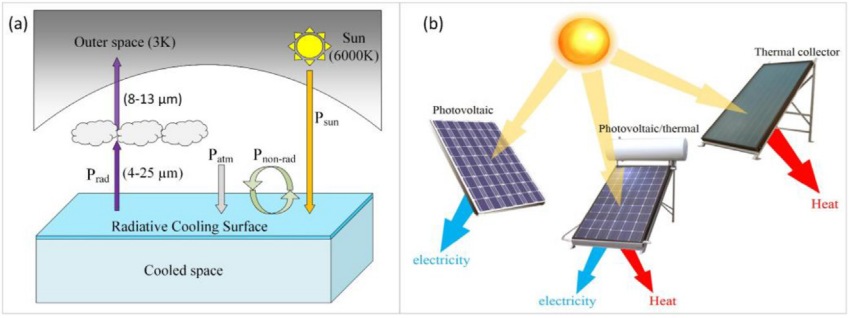


Fig. 1. **(a)** Basic working principles and energy flow for an RC system. P_{rad} is the emitted thermal radiation, P_{atm} is the absorbed atmospheric radiation, $P_{non\text{-}rad}$ is non-radiative heat exchange with ambient and P_{sun} is absorbed solar radiation. **(b)** Different types of SE harvesting systems. Reproduced with permission from Ref. [18].

clothes, cryogenic cooling, fluid cooling panels) [10–12]. Until recent past, RC could only be harnessed during nighttime, as the high solar gain during daytime required the RC materials to be highly reflective in the solar radiation band (i.e., 0.3–3 μm) and highly emissive in AW (8–13 μm). With the development in material science and manufacturing techniques, the stringent spectral requirements have now been achieved using novel materials (i.e., nanoparticle-based [13], photonic structures [14], metamaterials [15], etc.). However, the need for such materials has made daytime RC systems to be considerably more expensive than nighttime RC systems that made use of naturally occurring materials or simple polymers [16]. Other challenges faced by RC systems, which limit their wider application include low net cooling power gain (40–80 W/m^2) and long energy payback time [4,17].

Switching from fossil fuels to renewable energy is the need of the hour, and in this context, solar energy (SE) is most widely studied because of its abundance and potential to meet the global energy demand. As shown in Fig. 1b, SE harvesting systems can be broadly categorized into three main types, i.e., photovoltaic (PV), photovoltaic thermal (PVT), and thermal collector systems [18]. Despite developments over the years, SE systems face some critical challenges that need to be considered and resolved. Reduction in

power conversion efficiency because of elevated cell temperature is a major concern for solar PVs [19]. A 1 °C rise in the cell operating temperature results in a power conversion efficiency reduction of about 0.45% (for crystalline Si solar cells) [20], whereas the PV degradation rate doubles for every 10 °C rise in operating temperature [21]. As for conventional PVT systems, their efficiencies are low as they exhibit high MIR emissivities, which contribute to significant heat loss [22]. For solar thermal collectors, although the thermal efficiency is high, heating energy is the sole available output and that too only during daytime. Most importantly, as SE is intermittent and seasonally dependent, the optimum output gain is only achieved during peak sunshine hours, whereas the system stays idle during the remaining day [23–26].

It can be observed that individual RC and SE systems, both undergo similar challenges in terms of insufficient output energy gain, economic issues, and limited working time. It would be of particular interest if the functioning of individual systems could be integrated into a combined system that could provide enhanced output energy and function during both day and night. As most SE harvesting systems have a sky-facing flat plate structural design, which is similar to RC systems, it becomes practically feasible to integrate the two systems into a combined SE–RC system [27]. Within a single working unit, SE–RC systems produce a higher energy gain per unit area, and the total useful working time is increased because of system multifunctionality [28].

Several review papers regarding different subject matters within the RC research area have been published [4,6,7,17,23,29–40]. However, no review has so far comprehensively studied and analyzed the different types and developments, solely associated with combined SE–RC systems. Therefore, in this study, we have reviewed around 100 publications covering their system designs, analysis techniques, operating conditions, and output performances. To better understand the systems' working mechanism, simplified schematics have been shown. For comparative analysis, significant information from each study has been tabulated in the supplementary material. Challenges associated with the different system configurations and solutions considered are also discussed. Lastly, recommendations and suggestions for future researchers in this area have been presented.

2. Categorization of systems and theory

As shown in Fig. 2, SE harvesting systems are broadly categorized into three main types (i.e., PV, PVT, and thermal collector). When the RC technique is integrated with each of the SE systems, they can be further categorized into five types of SE–RC systems, based on their respective output gain and operating time (daytime and/or nighttime). In section 3.1, we discuss the first type that consists of studies that investigated daytime PV cooling to reduce solar cell operating temperature and improve the system's electrical efficiency. Section 3.2 discusses the second type of studies that considered nighttime RC in addition to the daytime PV cooling, whereas section 3.3 talks about the third type that considers both the daytime PV cooling and solar heating alongside the nighttime RC. The fourth type examines the systems that produce daytime solar heating and nighttime RC (section 3.4). Finally, in section 3.5, the fifth type of SE–RC systems explore simultaneous solar heating and RC during the daytime.

The inflow and outflow of energy for an RC surface as shown in Fig. 1a can be used to evaluate the net radiated cooling power as [6]:

$$P_{net,rad} = P_{rad} - P_{atm} - P_{sun} - P_{non,rad} \tag{1}$$

where P_{rad} is the thermal radiation power emitted from the RC surface, W/m^2; P_{atm} is the atmospheric radiation power absorbed by the RC surface, W/m^2; P_{sun} is the solar radiation power absorbed by the RC surface, W/m^2; $P_{non,rad}$ is the non-radiative (i.e., convective and conductive) heat transfer between the RC surface and ambient, W/m^2.

$$P_{rad} = 2\pi \int_0^{\frac{\pi}{2}} \int_0^{\infty} \varepsilon_s(\theta,\lambda) I_{bb}(\lambda, T_s) \sin(\theta)\cos(\theta) d\lambda d\theta \tag{2}$$

where $\varepsilon_s(\theta,\lambda)$ is the spectral directional emissivity of the RC surface, $I_{bb}(\lambda, T_s)$ is the blackbody spectral radiation intensity at T_s, W/(m^2·μm·sr); T_s is the temperature of the RC surface, °K; λ is the wavelength, μm.

$$P_{atm} = 2\pi \int_0^{\frac{\pi}{2}} \int_0^{\infty} \alpha_s(\theta,\lambda) \varepsilon_{atm}(\theta,\lambda) I_{bb}(\lambda, T_{amb}) \sin(\theta)\cos(\theta) d\lambda d\theta \tag{3}$$

where $\alpha_s(\theta,\lambda)$ is the spectral directional absorptivity of the RC surface, which is similar to the emissivity according to Kirchhoff's radiation law [41]; $\varepsilon_{atm}(\theta,\lambda)$ spectral directional emissivity of the atmosphere; $I_{bb}(\lambda, T_{amb})$ is the blackbody spectral radiation intensity at T_{amb}, W/(m^2·μm·sr); T_{amb} is the ambient temperature near the ground.

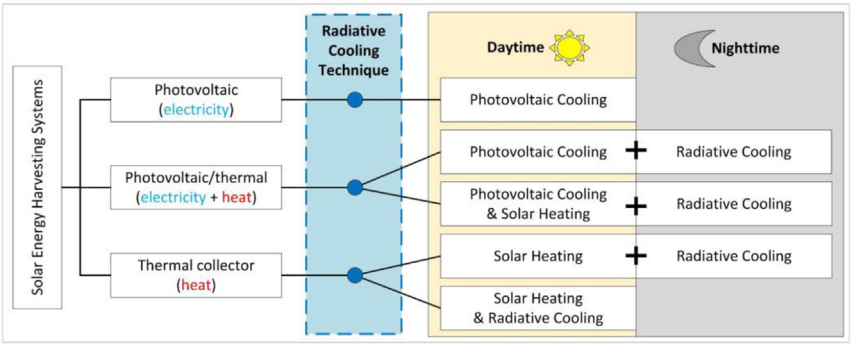


Fig. 2. Categorization of combined SE–RC systems based on their operating function and working time.

$$P_{sun} = G_{sun}\frac{\int_0^\infty \alpha_s(\theta_{sun},\lambda) I_{AM1.5}(\lambda) d\lambda}{\int_0^\infty I_{AM1.5}(\lambda) d\lambda} \quad (4)$$

where G_{sun} is the solar radiation, W/m^2;θ_{sun} is the angle between the normal direction of RC surface and sun that is assumed as fixed; $I_{AM1.5}(\lambda)$ is the AM 1.5 spectral distribution of solar radiation.

$$P_{non-rad} = h_{conv}(T_{amb} - T_s) \quad (5)$$

where h_{conv} is the convective heat transfer coefficient between the ambient and RC surface, W/(m^2·K). h_{conv} is a function of the velocity of wind (v_{wind}) and increases with it. When T_{amb} is higher than T_s, $P_{non\text{-}rad}$ is positive and reduces the net cooling effect, when T_{amb} is lower than T_s, $P_{non\text{-}rad}$ is negative and so it further enhances the net cooling effect.

It should be noted that in this review, the term 'atmospheric conditions,' refers to the combined effect of P_{sun}, P_{atm}, and $P_{non,rad}$. P_{sun} is primarily dependent on G_{sun}. P_{atm} depends mainly on T_{amb}, and ε_{atm}, which is related to RH or TWC. $P_{non,rad}$ depends on h_{conv}, which is related to v_{wind}. Therefore, lower numerical values for G_{sun}, $T_{amb,}$ RH/TWC, and v_{wind} would increase $P_{net,rad}$ and support RC, whereas higher values would reduce $P_{net,rad}$ and reduces the RC gain. For a more detailed understanding of the different variables involved to evaluate the overall RC effect, refer to Ref. [4].

For a solar PV system, the PV operating temperature (T_{PV}) is related to the electrical power generated (P_{elec}) as:

$$P_{elec} = G_{sun}\eta_{ref}\left[1 - \beta\left(T_{PV} - T_{ref}\right)\right] \quad (6)$$

where η_{ref} is the PV reference efficiency under standard test conditions; β is the PV temperature coefficient (0.0045 °C^{-1} for Si solar cells); and T_{ref} is the reference temperature (25 °C). From the relation, it can be observed that the lower the T_{PV}, the higher will be the η_{elec} of the system.

For a solar thermal collector, the thermal power (P_{ther}) harnessed by the system is evaluated as:

$$P_{ther} = \dot{m}c_p(T_{out} - T_{in}) \quad (7)$$

where $\dot{m}$ is the mass flow rate of the heat transfer fluid, kg/s; c_p is the specific heat capacity, J/(kg·K); and T_{in} and T_{out} are the inlet and outlet temperatures of the fluid in the collector, respectively.

3. Solar energy–radiative cooling systems

3.1. Daytime photovoltaic cooling

The reduction in power conversion efficiency because of the rise in PV operating temperature necessitates the importance of solar cell cooling [20,21]. Of the numerous PV cooling techniques (i.e., heat pipe, airflow, water flow, phase change materials, microchannel, liquid immersion/submerging, etc.), the recently introduced RC of solar cells has received increased interest as it is cheap, passive, and lightweight [42,43]. As shown in Fig. 3a, the radiative cooler (RCer) is generally placed on top of the solar cell and is designed to have two main radiative properties. Firstly, it should be highly transmissive in the solar band (0.3–4 μm) so that maximum sunlight is transmitted to the solar cell underneath to generate electricity. Secondly, the cooler should be highly emissive in the MIR band (4–25 μm) so that maximum heat is emitted out to reduce the solar cell temperature. It should be noted that thermal radiation only within the AW (8–13 μm) is released into outer space, whereas the remaining thermal heat is dumped into the local atmosphere.

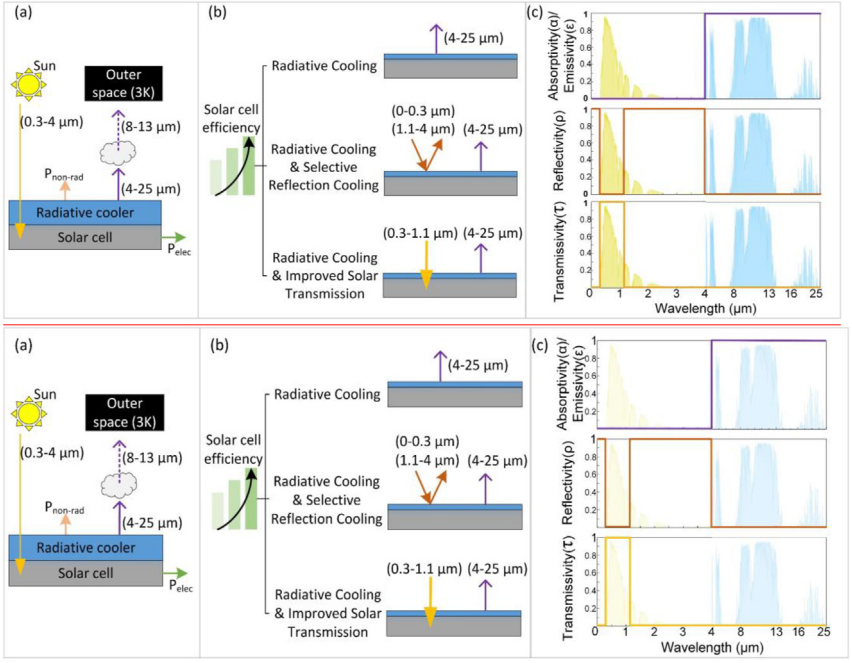


Fig. 3. **(a)** Schematic of daytime RC of a solar cell. **(b)** Categorization of studies that have investigated RC of solar cells. **(c)** Ideal radiative properties for an RCer placed above solar cells, vs. wavelength. Solar irradiance (light yellow) and atmospheric transmittance (light blue) are plotted as a reference.

RC can be categorized into above-ambient or below-ambient depending on the specific application need. As solar cells operate at a much higher temperature than T_{amb}, a broadband cooler (i.e., highly emissive in 4–25 μm) is preferred in contrast to a narrowband cooler (i.e., highly emissive in 8–13 μm), because it has larger cooling power for temperatures above T_{amb} [4,44].

In this review, we have considered silicon (Si) solar cells as a reference, which have a bandgap energy of 1.12 eV and a corresponding bandgap wavelength of 1.1 μm [45]. Photons below this wavelength create electron–hole pairs and generate electricity, whereas photons above the wavelength do not have enough energy and only add to PV heating. Some studies have investigated RCers, which not only have high emissivity but also reflect photons that do not contribute to electricity generation, using the technique of selective reflection. Such studies, which consider combined RC and selective reflection cooling (SRC) of solar cells, have been discussed in section 3.1.2.

The commonly used glass cover in PV modules causes reflection losses as it reflects some of the effective photons that could be used for electricity generation. Therefore, some studies have worked on designing RCers, which not only have high emissivity but also improve the solar transmission into the solar cell below. Studies that have investigated combined RC and improved solar transmission of solar cells have been discussed in section 3.1.3. Accordingly, the available literature on daytime PV cooling using the RC technique can be categorized into three types (Fig. 3b).

The ideal radiative properties for an RCer placed above the solar cell are shown in Fig. 3c. Firstly, high emissivity in 4–25 μm, to ensure maximum heat removal. Secondly, high reflectivity for wavelengths <0.3 μm (detrimental UV) and 1.1–4 μm, to reflect the photons that cannot generate electricity. Thirdly, high transmissivity in 0.3–1.1 μm, to ensure that maximum electricity-generating photons are received by the solar cell.

3.1.1. Radiative cooling

In this section, we discuss the studies that investigated RC of non-concentrated PVs and concentrated PVs.

3.1.1.1. Non-concentrated photovoltaics. Using a microphotonic technique, Zhu et al. [42] conducted the initial study and found that a transparent pyramid-shaped silica (SiO_2) photonic RCer was able to reduce the cell temperature by 17.6 °C as compared to a bare Si solar cell (Fig. 4a). In another study [46], the same group used the photolithography technique [47] to fabricate a photonic RCer by engraving air holes into a silica wafer as shown in Fig. 4b. In addition to the improved thermal emittance, the cooler slightly enhanced the sunlight absorption of the module. A temperature reduction of 13 °C (with wind-cover) and 5.2 °C (without wind-cover) was observed compared to a cell without RC. SiO_2 was used because of its high solar transmission and high IR emission properties.

In contrast to the aforementioned RC of flat-plate Si solar cells, some researchers investigated RC of other types of solar cells by changing the cell material and geometric shape. As shown in Fig. 4c, Wu and Povinelli [48] simulated the temperature reduction in nanowire-shaped GaAs solar cells for three different cases (i.e., planar-shaped cell, nanowire-shaped cell, and nanowire-shaped benzecyclobutene [BCB] coated cell). The BCB coating enhanced the thermal emissivity, which increased the RC power by 2.2 times and resulted in a 7 °C temperature drop compared to the planar-shaped solar cell. It was also found that nanowire structured cells showed higher thermal emissivity than conventionally used planar structured cells.

In an attempt to surpass the cell efficiency limit introduced by Shockley and Queisser [49], Safi and Munday [43] studied RC in GaAs and Si solar cells, for both terrestrial (narrowband emitter in AW) and extraterrestrial environment (broadband emitter and no P_{atm}). As shown in Fig. 4d, the proposed cell with an RCer increased the cell efficiency by 0.87% compared to a cell operating at 335 °*K*, which is the usual operating temperature of solar cells in terrestrial conditions. For the extraterrestrial environment, a higher efficiency improvement of 0.4%–2.6% was observed. In another study [50], the same group studied RC for a GaAs solar cell and found that at standard operating conditions, a temperature drop of 18 °C could be achieved when compared to a cell without RC (Fig. 4e).

Lee et al. [51] studied RC in flexible thin-film solar cells using pyramid-shaped polydimethylsiloxane (PDMS) RCer that had near-ideal emissivity. Thermal simulations showed that cell temperature could be reduced by 11 °C (for organic), 12 °C (for perovskite), and 16 °C (for c-Si) solar cells, as compared to the conventional solar cells that used a planar shaped polyethylene terephthalate (PET) substrate (Fig. 4f). Although PDMS has high flexibility, easy-fabrication, and costs low, which make it a suitable material for RC of thin-film solar cells, its durability was not studied. Using a similar approach, Banik et al. [52] studied RC in thin-film CIGS solar

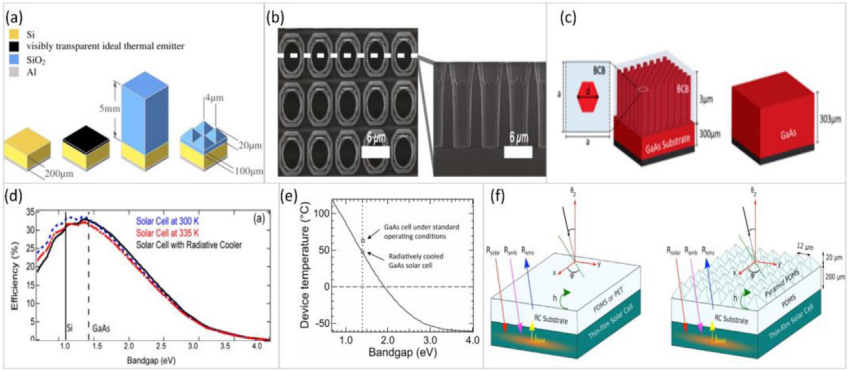


Fig. 4. **(a)** Schematic of a pyramid-shaped photonic RCer on top of an Si cell (on most right) compared to a bare Si cell (on most left). Reproduced with permission from Ref. [42]. **(b)** Microscopic image of the top and side views of the photonic crystal structure fabricated by engraving air holes into fused silica wafer. Reproduced with permission from Ref. [46]. **(c)** Nanowire-shaped GaAs cell, embedded in BCB (on left), and a planar-shaped GaAs cell (on right). The inset is the magnified top view of one-unit cell. Reproduced with permission from Ref. [48]. **(d)** Efficiency comparison of the proposed solar cell (black line), cell operating at room temperature of 300 K (dashed blue line), and cell operating at a typical elevated temperature of 335 K (red line), vs. bandgap energy. Vertical lines indicate the bandgaps of GaAs and Si cells, respectively. Reproduced with permission from Ref. [43]. **(e)** Operating temperature vs. bandgap energy of radiatively cooled GaAs solar cell compared to a cell without RC. Reproduced with permission from Ref. [50]. **(f)** Flexible thin-film solar cells (dark green) with planar-shaped (on left) and pyramid-shaped (on right) RC substrate. Reproduced with permission from Ref. [51].

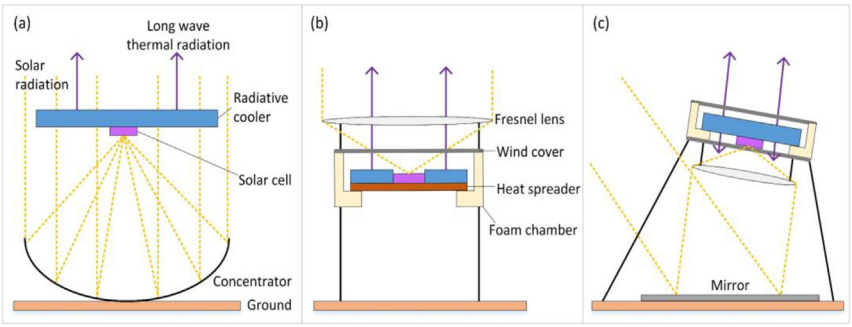


Fig. 5. **(a)** Schematic of the theoretical concept of a radiatively cooled CPV system. The system consists of a parabolic reflector (black) that concentrates sunlight (dashed yellow lines) onto the solar cell (magenta). The RCer (blue) placed on the backside of the cell has high solar transmission (so that the concentrator collects sunlight) and radiates heat on both sides (to cool the solar cell). Redrawn from Ref. [54]. **(b)** Schematic of the experimental setup for outdoor field testing. The Fresnel lens (light grey) concentrates sunlight onto the solar cell. The wind cover (dark grey) limits convection while the heat spreader (orange) ensures heat is evenly conducted from the cell to the cooler. Redrawn from Ref. [55]. **(c)** Schematic of the experimental setup for outdoor field testing. A mirror (grey) is placed on the ground to reflect sunlight normally to the Fresnel lens. Thermal radiation is emitted from both sides of the RCer. Redrawn from Ref. [44].

cells that are usually used in space applications. As CIGS solar cells have a very low thermal emissivity, using a polymeric coating (silicon oxycarbonitride) it was increased from 0.3 to 0.72, which led to a maximum cell temperature reduction of 30 °C under optimum conditions.

3.1.1.2. Concentrated photovoltaics. In contrast to the RC of non-concentrated PVs, in this section, we look into the RC of concentrated PVs (CPV), as they operate at much higher temperatures because of sunlight concentration and therefore have higher potential for cell temperature reduction using RC [53]. Sun et al. [54] introduced the concept and simulated the performance using a multilayered soda-lime glass with a porous layer on top as the RCer, which also reduced solar reflection (Fig. 5a). The cell temperature was decreased by 14 °C as compared to the conventional copper cooler used in CPV systems. As shown in Fig. 5b, the concept was then experimentally demonstrated by Zhou et al. [55] using an RCer that comprised of soda-lime glass for high MIR emissivity and aluminum back reflector for sunlight reflection and keep the RCer cool. The solar cell temperature was reduced by 10 °C, which corresponds to a V_{oc} increment of 5.7% and predicts a PV lifetime increment of 40%. Recently, Wang et al. [44] carried out multiple experiments and simulations to study RC using similar soda-lime glass-based RCers. They found that RC performance varies significantly with changes in cooler design, wind speed, and heat load. For optimum operating conditions, the cell temperature was reduced by 36 °C, (i.e., 27% rise in V_{oc}) compared to a system with no RC. It should be noted that a very large temperature reduction was observed because the experimental chamber was sealed (no convection cooling) and RC was achieved from both the front and the backside of the RCer (Fig. 5c). In the presence of convection cooling, the multilayered photonic RCer developed by Li et al. [56] was simulated in a CPV system. The cooler reduced the cell temperature by 15.4 °C when the h_{conv} at module backside was 30 W/(m^2·K), and by 6.2 °C when 200 W/(m^2·K). Considerable temperature reduction even with large h_{conv} for the studies discussed indicates the higher potential of using RC in CPV systems, compared to non-concentrated PV systems.

Different types of materials have been investigated for the design of RCers. These include polymer based, photonic based, and nanoparticle based. Polymers like PDMS and PET have high solar transparency in solar band and high emissivities in IR band. Although polymers are easy to manufacture, their life expectancy and weather durability are major concerns. As for photonic RCers, they can be further categorized into multilayered film stacks, patterned structures, or a combination of both. Multilayered film stacks [56] consist of alternating layers of material with different dielectric constants and varying thickness, to get the desired absorptivity/emissivity. As for patterned structures (i.e., pyramids [42], air holes [46], gratings [57], crystals, etc.), they provide a high degree of freedom for tailoring the spectral radiative properties according to requirements. Nanoparticle-based RCers (or metamaterials) have a precise geometry, arrangement, and design that manipulates electromagnetic waves and thus achieve spectral properties beyond those of naturally available materials.

3.1.2. Radiative cooling and selective reflection cooling

The studies discussed in the previous sections investigated PV cooling using only RC. In this section, we analyze the studies that investigated combined RC and SRC of solar cells. Initially, Li et al. [56] simulated a multilayered (Al_2O_3, SiN, TiO_2) photonic RCer placed atop an Si solar cell, which not only increased the thermal emissivity but also reflected the useless sub-bandgap photons, and reduced the cell temperature by 5.7 °C (Fig. 6a). To evaluate the respective share of each cooling technique (i.e., RC and SRC) in the overall temperature reduction, Sun et al. [58] compared both techniques using simulations on different types of solar cells (i.e., GaAs, CdTe, CIGS, Si) each with its absorptivity profile shown in Fig. 6b. Results showed SRC to be more effective than RC, whereas the combined RC and SRC cooling gave the highest temperature reduction (Fig. 6c). A similar comparative study was performed by Li et al. [59]; however, they found RC to be more effective than SRC (Fig. 6d). It was observed that the PV module efficiency was improved by 0.98% (for SRC), 2.4% (for RC), and 4.55% (for combined cooling). The study also predicted the power generation gain and economic viability of the cooling techniques using parameters like integrated energy-saving efficiency and levelized cost of electricity [60].

As previous studies ignored major intrinsic processes affected by PV temperature reduction, Perrakis et al. [61] developed an experimentally verified detailed electrical–thermal model in their study. They also considered the impact of UV reflection in addition to sub-bandgap reflection and RC. It was found that when all cooling approaches (UV reflection, sub-bandgap reflection, RC) were combined, the PV temperature was reduced by around 12 °C (Fig. 6e). In contrast to the simplified 1D models considered by previous studies, An et al. [57] simulated multidimensional models and considered the coupled effect between RC, carrier

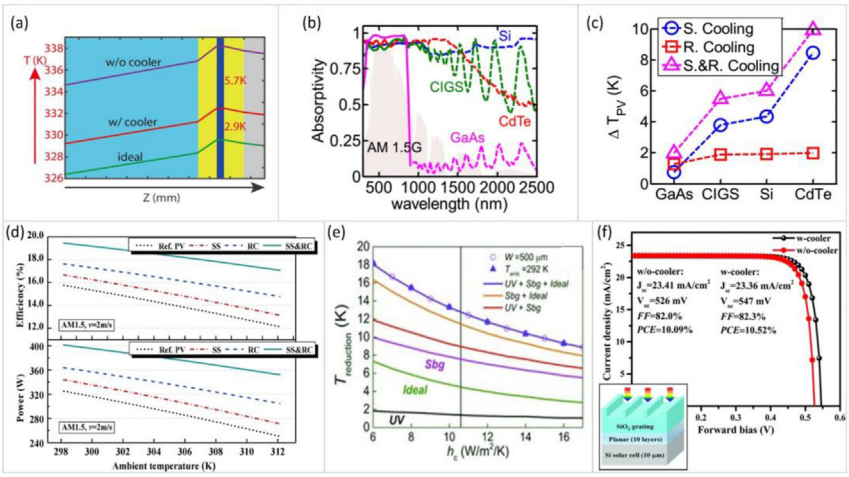


Fig. 6. **(a)** Simulated temperature distribution across a PV module without a cooler (purple line), with proposed cooler (red line), and with an ideal cooler (green line). Reproduced with permission from Ref. [56]. **(b)** Measured absorptivity spectra for different solar cell materials vs. photon wavelength (solid lines: above bandgap photons; dashed lines: below bandgap photons). **(c)** PV temperature reduction using different cooling techniques for different types of solar cells (red squares: individual RC; blue circles: individual SRC; pink triangles: combined RC and SRC). Figures b and c reproduced with permission from Ref. [58]. **(d)** Efficiency and power output vs. T_{amb}. RC (dashed blue) is more effective than SRC (double dashed red), whereas combined RC and SRC (turquoise) is most effective. Reproduced with permission from Ref. [59]. **(e)** PV temperature reduction using different cooling techniques vs. h_{conv}. UV reflection (black), sub-bandgap reflection (magenta), ideal RC (green), combined UV and sub-bandgap reflection (red), combined UV and sub-bandgap reflection along with ideal RC (purple). Reproduced with permission from Ref. [61]. **(f)** Effect of RC on the solar cell IV curve (black: with cooler; red: without cooler). Inset shows the RCer that consists of a multilayer thin-film stack (turquoise) and an SiO_2 grating (light blue), placed above an Si cell (grey). Reproduced with permission from Ref. [57].

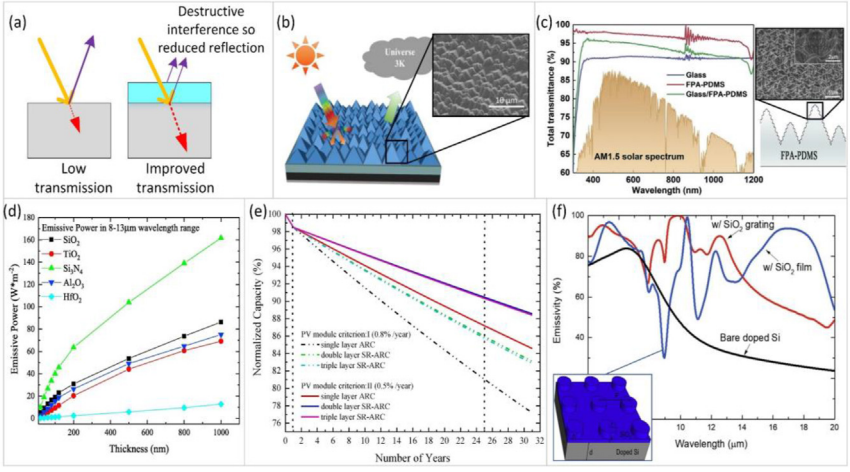


Fig. 7. **(a)** A PV module (grey) without ARC having low transmission (on left), compared to a PV module with ARC (turquoise) having improved transmission (on right). Destructive interference between waves reflected from ARC and PV module reduces reflection losses and improves the transmission of light. **(b)** Schematic of the sol–gel imprinted textured glass used in PV modules as cover for improving the solar transmission and radiative emission. Reproduced with permission from Ref. [65]. **(c)** The transmittance of bare glass (purple line), proposed FPA-PDMS (red line), and FPA-PDMS on glass (green line), vs. wavelength (on left). Schematic of the concept and SEM image of the flexible photonic structure inspired from silver ant hair (on right). Reproduced with permission from Ref. [67]. **(d)** Variation in emissive power vs. thickness for different ARC materials. Si_3N_4 (green) has the highest emissive power. Reproduced with permission from Ref. [68]. **(e)** PV module generation capacity against time for different ARC layer combinations at degradation rate of 0.5%/year (solid lines) and 0.8%/year (dashed lines) at a T_{amb} of 30 °C. Double-layered ARC (solid purple, dashed green) has the smallest degradation rate. Reproduced with permission from Ref. [69]. **(f)** Simulated thermal emissivity for each case, which consists of a bare solar cell (black line), solar cell with plain silica cooler (blue line), and solar cell with silica micrograting cooler (red line). Inset shows schematic of silica micrograting (dark blue) on top of Si cell (grey). Reproduced with permission from Ref. [70].

thermodynamics, and electrodynamics in solar cells. As shown in Fig. 6f, the effect of the RCer on the performance of I_{sc}, V_{oc}, fill factor, and power conversion efficiency was studied. The RCer considered was a multilayered film stack with an SiO_2 grating atop of Si solar cell. Temperature reduction of 10 °C and cell efficiency rise of 0.43% were observed. SiO_2 was selected because of its high extinction coefficient in IR band and high transmission in solar band. As for the grating structure, it was used to suppress the reduction in emissivity, which occurs for a planar structure at 9 and 21 μm due to impedance mismatch between air and SiO_2.

3.1.3. Radiative cooling and improved solar transmission

It is generally known that solar PVs have high surface reflection (~30% for c-Si cells) [62]. Texturing the top surface using an anti-reflective coating (ARC) can significantly reduce reflection losses and improve cell efficiency. As shown in Fig. 7a, an ARC is a thin layer of dielectric material, with a specific thickness [63]. The waves reflected from the ARC top surface are put out of phase with the waves reflected from the solar cell surface, causing destructive interference between them, which results in reduced reflection. In this section, we discuss the studies that alongside RC have

investigated ways to improve sunlight transmission into the cells underneath the RCer.

Slauch et al. [64] designed a photonic mirror, with common dielectric materials SiO_2, Si_3N_4, TiO_2, and Al_2O_3, which not only reflected sub-bandgap photons but also acted as an ARC, and increased the PV energy output by 3.6% over a year. The optimum position to place the photonic mirror in the module was also investigated and was found to be above the glass cover. As shown in Fig. 7b, Lu et al. [65] fabricated a pyramid textured glass cover, which had a thermal emissivity and solar transmissivity of about 94%. Textured covers reduce solar reflection due to the multiple rebound of incident light as compared to a conventional flat glass cover [66]. Results showed that by applying the textured glass cover, the PV modules I_{sc} and power conversion efficiency were increased by about 5.12% and 3.13%, respectively. The sol-gel imprinted fabrication process used in this study was scalable, durable, and energy-efficient as compared to the photolithography process used by Ref. [46].

Inspired by the hair of silver ants, which show high radiative heat dissipation, Lin et al. [67] fabricated a flexible photonic architecture (FPA) with a corrugated triangular cross-section, which had high MIR emittance and high solar transmittance. Fig. 7c shows the improved transmittance of the photonic structure, which was denoted as FPA-PDMS in the study. Bare glass (in blue), FPA-PDMS attached to glass (in green), and simple FPA-PDMS (in red), had an average transmittance of 90.34%, 92.70%, and 96.72%, respectively. When the structure was electrostatically attached on top of glass-encapsulated solar cells, the power conversion efficiency was increased by 3.23% (for c-Si) and 6.75% (for perovskite) solar cells, as compared to conventional glass-encapsulated solar cells. The use of a bio-inspired RCer was also studied by Jeong et al. [8].

Kumar and Chowdhury [68] studied the impact of applying different ARC material films (SiO_2, Al_2O_3, Si_3N_4, HfO_2, TiO_2) on top of Si solar cells. The Si_3N_4 coating showed the best results as it gave the highest emissivity in AW and a considerable rise in solar transmission. It was also found that the emissive power of the coatings improved with the increase in coating thickness, as shown in Fig. 7d. The same group then used Si_3N_4 in single-layer and multilayer configurations to study its performance on a c-Si solar cell [69]. A temperature reduction of 5.4 °C (for the triple-layer) and 5.6 °C (for the double-layer) was observed compared to a single-layer coating. The study also investigated the effect of RC on solar cell power degradation rate against the years of field operation. As shown in Fig. 7e, it can be seen that a double-layered ARC had a smaller degradation rate compared to single- and triple-layered coatings. Long et al. [70] fabricated silica micrograting for antireflection and RC of Si solar cells (Fig. 7f). Three cases were analyzed in the study (i.e., a solar cell with silica micrograting RCer, a solar cell with a plain silica RCer, and a bare solar cell).

3.1.4. Other photovoltaic/photothermic cooling applications

In contrast to the integration of RC with solar PVs, some studies have investigated RC with other PV/photothermic applications such as thermophotovoltaics (TPVs) [71], PV-thermoelectric [72], and concentrated solar power (CSP) [73,74], which are discussed below.

3.1.4.1. Thermophotovoltaic system. TPV systems use PV cells to harvest electricity from thermal radiations at desired wavelengths. The ability to control the emitting wavelengths using an appropriately designed thermal emitter allows TPV systems higher power density as compared to solar PV systems [75]. However, as the emitter works at extremely high temperatures (>800 °C), it creates a severe thermal management challenge for the PV cells. Therefore, removing heat from PV cells in a TPV system using RC is a possible solution.

Zhou et al. [71] conducted the first simulation study on integrating RC in a TPV system, as shown in Fig. 8a. After receiving thermal energy from a heat source, the thermal emitter emits photons at certain wavelengths that are effectively absorbed by the PV diode facing it. The RCer is exposed to the ambient and emits heat, eventually reducing the PV operating temperature. It was found that using a photonic RCer, the TPV system efficiency increased by 18% as compared to a system without RC. Also, a significant increment in PV cell lifetime was predicted.

3.1.4.2. Photovoltaic-thermoelectric system. As solar radiation below the PV bandgap generates waste heat, by combining a thermoelectric module to a PV module, it can be directed to the hot side of the thermoelectric module to generate electricity by the thermoelectric effect [76]. The impact of RC to enhance heat transfer from the cold side of a thermoelectric generator was studied by Liu et al. [77]. In their subsequent study [72], the effect of RC was investigated by integrating it into a hybrid PV-thermoelectric system (Fig. 8b) model developed in an earlier study [78]. Different cover coatings were used as the RCer and a considerable rise in power generation was achieved. The ability to generate some electricity (~25 mW/m^2) only during the night, by RC of the cold side of a thermoelectric generator was also investigated [79]. Furthermore, integration of RC to a PV-powered thermoelectric cooler was recently proposed, where the system could provide cooling during both day and night [80].

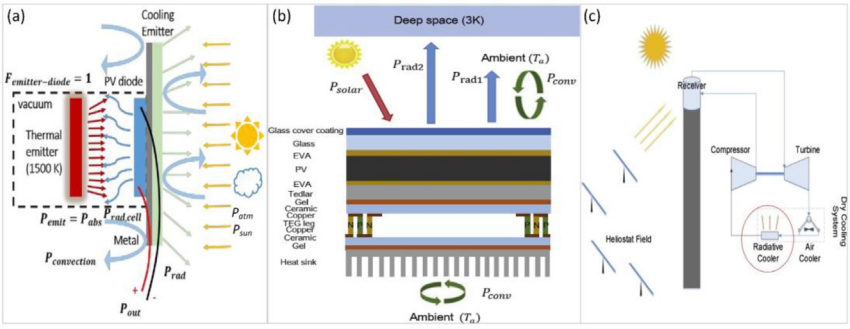


Fig. 8. (a) Schematic of a radiatively cooled TPV system. The thermal emitter (red) and the PV diode (blue) are enclosed in a vacuum chamber (dashed black lines). The cooling emitter (green) and part of the heat spreader (grey) are exposed to ambient. Reproduced with permission from Ref. [71]. (b) Schematic of a radiatively cooled PV-thermoelectric system. Glass cover coating (dark-blue) works as the RCer and emits heat from the PV (dark brown) into outer space. Reproduced with permission from Ref. [72]. (c) Schematic of a CSP power plant having an RCer (in red circle) integrated into a conventional air cooler. Reproduced with permission from Ref. [73].

3.1.4.3. Concentrated solar power system. CSP systems generate electricity using mirrors and lenses that concentrate sunlight onto a receiver. The receiver is a heat-carrying medium such as water, which gets converted into steam that drives the turbine connected to a power generator to generate electricity [81]. CSP power plants usually make use of active cooling techniques (i.e., air/water cooling). This has some challenges mainly because water is a scarce resource and the parasitic load for circulating the fluid, significantly reduces the plant's efficiency [82]. As RC is resource-efficient and can provide complementary cooling, RC in CSP power plants is another application area that was considered. Also, as most CSP power plants are installed in regions with a desert climate, the atmospheric conditions are very suitable for RC as relative humidity is considerably low.

Zeyghami and Khalili [73] investigated the use of daytime RC as a complementary cooling method to further enhance the performance of an air-cooled CSP power plant. The RCer was integrated into the cycle after the air cooler, as shown in Fig. 8c, to further reduce the working fluid temperature before the compressor inlet. It was found that RC reduced the cooling load on the air cooler, which reduced the parasitic loads and improved the cycle performance. A similar study was conducted by Dyreson and Miller [74] who simulated the impact of RC in a CSP power plant using a detailed heat transfer model. It was found that 90% of the annual cooling requirements could be achieved using an RC system of the same size as the solar collector field. Apart from CSP, the RC technique can be investigated to provide supplementary cooling to any thermal power plant where water usage is a concern.

3.1.5. Comparison with glass-covered photovoltaics

It should be noted that most studies discussed previously have compared the performance of PV modules with highly emissive RCers against unencapsulated (bare Si) PV modules that have relatively very low emissivity (0.45–0.65). As a result, large reductions in PV operating temperatures were reported and RC was reviewed as an impressive cooling technique for solar cells. However, these results seem to be overestimated as commercial solar cells are usually encapsulated with glass (SiO_2), which naturally has a high thermal emissivity and therefore can provide RC even without using any additional RCers on top. Riverola et al. [83] investigated the emissivity of commercial Si solar cells using a comprehensive model that considered the complete cell structure and texture, which was ignored by Refs. [14,42]. It was found that both the unencapsulated and encapsulated solar cells were inherently good thermal emitters with MIR emissivities of 0.80 and 0.90, respectively. Mellor et al. [22] also found that PVs, both with and without the glass cover, have considerably high emissivities, as shown in Fig. 9a. Therefore, it can be expected that under realistic atmospheric conditions, replacing the regular glass cover with a highly emissive RCer on top of the solar cells would not bring any considerable temperature reduction.

Some studies have comparatively analyzed the performance of glass encapsulated cells to those with an ideal ($\varepsilon = 1$) RCer op top. Gentle and Smith [84] carried first such comparative study and found insignificant PV temperature reduction. Similarly, Sato and Yamada [85] simulated three cases (ideal broadband emitter, ideal narrowband emitter, and conventional glass cover), and found out that using an ideal emitter reduced the PV temperature by just 1 °C compared to the glass cover (Fig. 9b). Zhao et al. [86] experimentally demonstrated that enhancing the RC ability by applying a thin PDMS layer on a glass encapsulated solar cell had insignificant performance improvement. Following a similar approach, Ahmed et al. [87] studied the effect of enhanced RC in a PVT system considering an exergy analysis. As shown in Fig. 9c, the total efficiency of the system was improved by a mere 0.5%, throughout the day. Recently, Li et al. [88] carried outdoor field tests and found that emissivity enhancement of PV modules within the AW, using PDMS coating, could not reduce the cell temperature by even 1 °C. Thus, it is evident that the substitution of glass is not considerably useful for sole RC in PVs.

3.2. Daytime photovoltaic cooling and nighttime radiative cooling

In this section, we discuss the second type of SE–RC systems, which not only consider daytime RC of PVs but also harness cooling energy from RC during nighttime. The cooling energy is first extracted by the absorber/emitter (a/e) plate that is placed underneath the solar cells as in a typical PVT system configuration. It is then transferred to the heat transfer fluid (i.e., air, water, etc.) flowing underneath the a/e plate and is used accordingly. It should be noted that the words emitter and RCer have been used interchangeably in this paper.

The PVT systems considered by the studies discussed in this section can be classified into four main types as shown in Fig. 10a. Type A has no wind cover, whereas type B consists of a glass wind cover that also prevents the PVs from environmental damage (i.e., rain or dust). Type C has a glass cover, whereas type D has a polyethylene (PE) film cover, each type with an air gap between the PVs and the cover. The air gap reduces heat loss to the ambient and hence improves thermal efficiency. However, the wind cover is not helpful for electrical efficiency as the photons reaching the solar cells are reduced because of the absorption and reflection losses of the cover [89]. From the transmittance profile of glass and PE, as shown in Fig. 10b, it can be seen that both materials have high

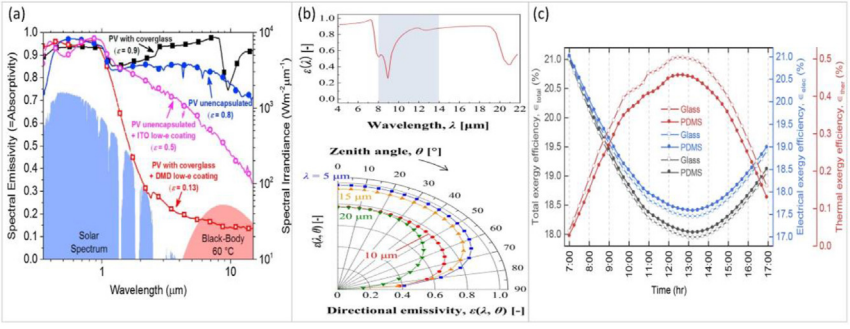


Fig. 9. **(a)** Experimentally measured absorptivity/emissivity spectra of PV modules (black squares: with glass cover = 0.9; blue circles: without glass cover = 0.8). Reproduced with permission from Ref. [22]. **(b)** Spectral emissivity (top) and directional emissivity (bottom) for conventional glass cover case. Reproduced with permission from Ref. [85]. **(c)** Total exergy efficiency (black), electrical exergy efficiency (blue), and thermal exergy efficiency (red) for glass-covered (unfilled circles) and PDMS-coated glass-covered (filled circles) PVT modules during the daytime. Reproduced with permission from Ref. [87].

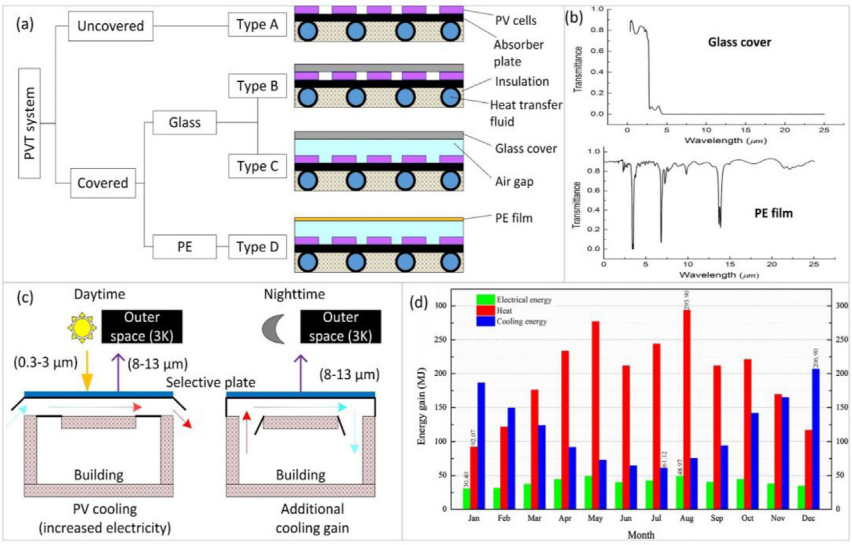


Fig. 10. **(a)** Classification (on left) and schematic (on right) of the daytime PV cooling and nighttime RC cases investigated in the literature, based on the type of wind cover used. **(b)** Transmittance profile vs. wavelength of a glass cover (top) and PE film (bottom). Glass is almost opaque in the MIR band, whereas PE is highly transmissive and so suitable for thermal wavelengths to escape through the system. Reproduced with permission from Ref. [99]. **(c)** The theoretical concept of a building-integrated daytime PV cooling and nighttime RC system. During daytime, the system provides increased electricity due to RC of solar cells, whereas during nighttime, the system can be used for space cooling of the building. Redrawn from Ref. [91]. **(d)** Simulated annual energy gain achieved from a daytime PV cooling and solar heating and nighttime RC system in Hefei (green: electrical energy; red: thermal energy; blue: cooling energy). Reproduced with permission from Ref. [98].

transmittance in the solar band, whereas in the MIR band, glass is opaque while PE is transparent. This property makes PE an appropriate material to be used as a wind cover to allow for the thermal radiative emission to the sky and therefore be used for RC. Although, removing the wind cover (i.e., Type A) would allow for radiative emission to the sky, which is suitable to RC, but the convective cooling from the wind would reduce the RC potential as studied by Liu et al. [90]. Note that type D (with the PE wind cover) may consist of either unencapsulated PVs or encapsulated PVs, for which the only difference is the MIR emissivity of PV top surface (~0.5 for unencapsulated and ~0.85 for encapsulated).

Eicker and Dalibard [11] conducted the first study on daytime PV cooling and nighttime RC systems, in which they tested the performance by integrating type A modules on a residential building in Madrid, and observed nighttime RC power gain of 60–65 W/m^2 from a weak long experimental testing. It should be noted that their system did not consider any spectrally dependent optical properties for RC. The nighttime thermal radiation to the sky was made possible by only removing the glazing from the PVT module.

With the development in material science, Zhao et al. [91] considered the spectrally dependent approach and proposed a selective plate (a combination of solar cells and RCer), with high absorptivity in the solar band (for electricity generation) and high emissivity in AW (for daytime and nighttime RC). As shown in Fig. 10c, the selective plate was integrated into a building's rooftop and a thermal model was developed to simulate the system's performance in Hefei. Later on, a hybrid module was manufactured to experimentally demonstrate this concept [92]. The RCer used was a transparent tedlar polyester tedlar layer that has high MIR emissivity (~0.94).

In another study, Zhao et al. [93] designed a photonic structure consisting of 2D photonic crystals and a 1D multilayered stack, placed above the Si solar cell. The multilayered stack provided sub-bandgap reflection, whereas the photonic crystals enhanced the thermal emittance of the structure. The thermal simulation showed that the photonic cooler improved electric power generation by 4.6% and cooling gain by 9.3% compared to the conventional glass-covered solar cell. Considering the inherently high emittance of glass, an experimental study was conducted to quantify the nighttime cooling gain that could be harnessed from a conventional glass-covered PV module [94]. It was observed that the PV temperature dropped by 12.7 °C below T_{amb}, which proved that the conventional PV module could indeed be used to harness nighttime RC.

3.3. Daytime photovoltaic cooling and solar heating and nighttime radiative cooling

In contrast to the previous section, we now discuss the studies that have considered both the RC of solar cells and the collection of thermal energy during daytime, whereas for nighttime, RC gain was harnessed. Initially, Bilbao and Sproul [95] investigated the performance of an uncovered PVT collector similar to the one studied by Eicker and Dalibard [11]. Later on, Sohel et al. [96] carried out a study to find the feasibility of nighttime RC in a building-integrated PVT system and found that nighttime RC was considerable in hot climatic conditions, whereas in colder regions, it was not significant.

Considering the inherently high MIR emissivity of glass, Ahmed et al. [87] conducted a performance comparison of a regular glass-covered PVT module and a spectrally enhanced PDMS-coated PVT module (type B). Results showed that by enhanced RC, the total exergy efficiency of the system was improved by 0.5% during the daytime, and about 4–7 W/m^2 of additional cooling gain was obtained at nighttime. Although an improvement, the potential gains in the system performance were not substantially large and it was concluded that substituting glass with high emissive materials would not be a viable option for commercial PVT systems.

Hu et al. [28] developed a steady-state mathematical model and conducted a sensitivity analysis to identify the key parameters of the system. In their subsequent study [97], an experimental setup was designed and tests were performed at various operation modes to validate the model developed previously. Later on, the previous experimental setup was improved, and a new study was conducted with a more detailed numerical model [98]. As shown in Fig. 10d, the annual performance of the system for different energy output

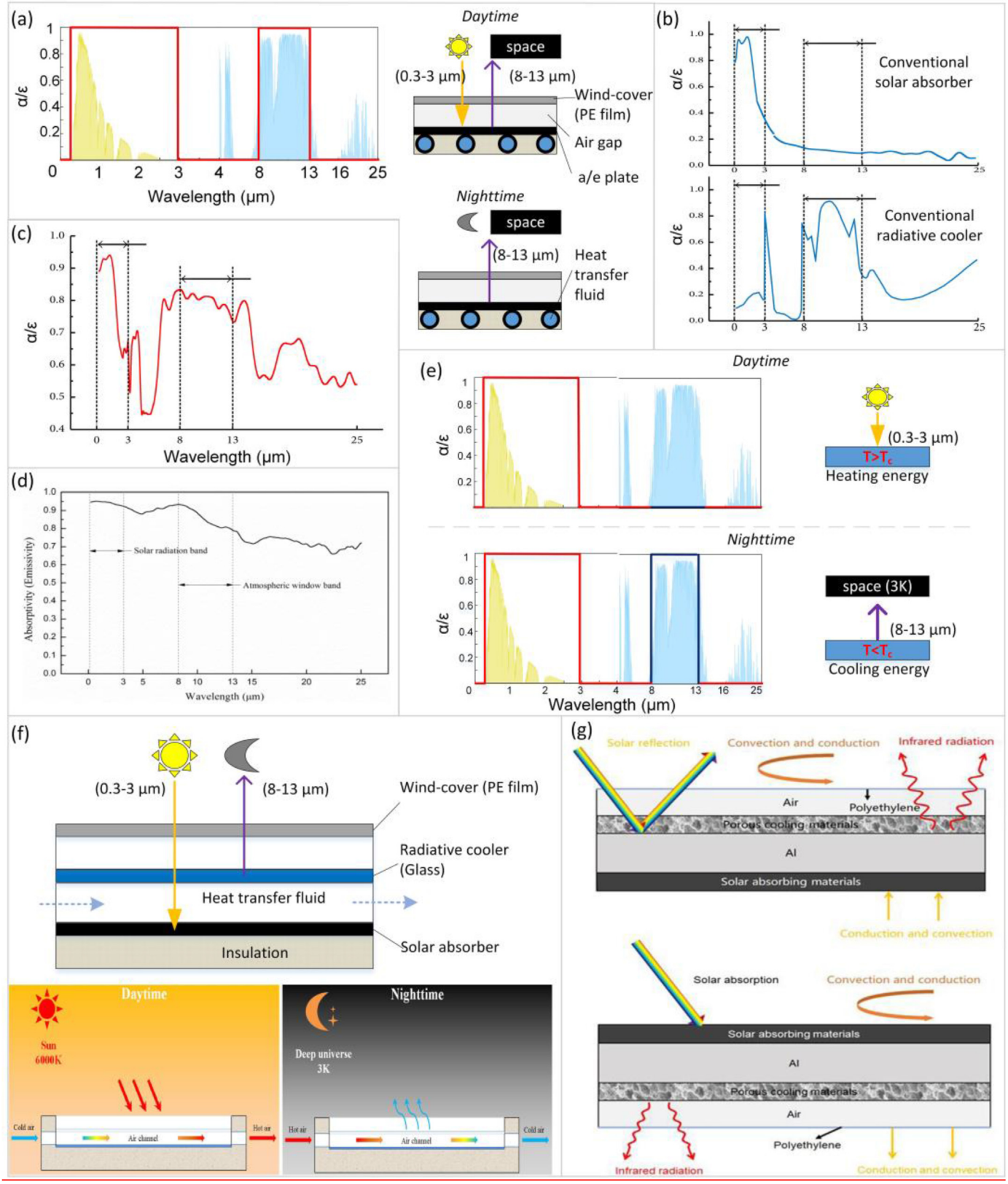


Fig. 11. **(a)** Ideal absorptivity/emissivity profile for the a/e plate in daytime solar heating and nighttime RC system. High absorptivity in the solar band (0.3–3 μm) ensures maximum heat is absorbed in, whereas a high emissivity in AW (8–13 μm) ensures maximum heat is emitted out. **(b)** Absorptivity/emissivity profile of a conventional solar absorbing plate (on top) and conventional RC plate (on bottom). Reproduced with permission from Ref. [103]. **(c)** Absorptivity/emissivity profile of the a/e plate designed in Ref. [103]. **(d)** Absorptivity/emissivity profile of black acrylic paint coating used in Ref. [106]. **(e)** Ideal absorptivity/emissivity profile and schematic of the proposed self-adaptive a/e plate during daytime (on top) and nighttime (on bottom) working mode. Redrawn from Ref. [110]. **(f)** Schematic of the proposed module (on top) consisting of a separate RCer (blue) placed above the absorber plate (black) with an air channel in between. PE film (grey) is used as the wind cover to transmit thermal radiation into outer space. The working mechanism of the proposed module during daytime (left bottom) and nighttime (right bottom). Reproduced with permission from Ref. [114]. **(g)** The module in cooling mode (on top) during summers and in heating mode (on bottom) during winters when turned upside down. Reproduced with permission from Ref. [116].

parameters (i.e., electricity, cooling, heating) was evaluated in Hefei, China.

3.4. Daytime solar heating and nighttime radiative cooling

The SE−RC system that provides solar heating during daytime and RC during nighttime have been discussed in this section. To have maximum heating gain at day and RC at night, the a/e plate of the combined system should have an ideal absorptivity/emissivity, as shown in Fig. 11a. High absorptivity in the solar band would provide a large heating gain, whereas high emissivity in AW would provide a large cooling gain. The low absorptivity/emissivity in other bands would reduce any heat loss during the day and cooling loss (i.e., heat gain) during the night.

In contrast to an ideal daytime solar heating and nighttime RC system (Fig. 11a), the radiative properties of the a/e plate of an individual solar absorber and RCer have been shown in Fig. 11b. It should be noted that a conventional solar absorber cannot be used for nighttime RC because of its low emissivity in AW. Similarly, a conventional RCer cannot be used for daytime solar heating because of its low absorptivity in the solar band. Studies that have investigated such systems with absorptivity/emissivity close to the ideal system are discussed below.

The capability of typical flat plate solar absorbers to be used for nighttime RC was first studied by Matsuta et al. [100]. They conducted a day-long experiment on a spectrally selective a/e plate (black copper plate coated with thin polyvinyl-difluoride film) and found that the plate had a lower thermal efficiency than conventional solar absorbers. This was because of the polymer film's high emissivity in AW, a property useful for nighttime RC but not suitable for thermal collectors. Following a similar approach, Erell and Etzion [10] studied the use of flat-plate solar absorbers for RC in buildings.

The feasibility of a desiccant-based daytime solar heating and nighttime RC system was investigated in hot arid Egypt by Ali [101]. Using a mathematical model, it was found that with a lower air mass flow rate, the outlet air temperature at nighttime could be reduced by about 5.5−7 °C below T_{amb}. Also, the outlet air would have an RH of below 40% because of the desiccant, making it suitable for air comfort applications. In contrast to previous studies that had considered a roof-mounted system, Yong et al. [102] experimentally investigated a wall-mounted daytime solar heating and nighttime RC system. It was found that the system produced a daily average heat collecting efficiency of 39% during winters and an average cooling capacity of 87 W/m^2 during summers.

Later on, Hu et al. [103] fabricated a spectrally selective a/e plate with a titanium-based solar absorber and a PET RCer. Fig. 11c shows the absorptivity/emissivity profile of the plate. Results showed that the plate had a heating efficiency of 76.8% relative to the conventional solar absorber. Using the same plate, the group then performed several studies such as an annual performance evaluation [104], theoretical comparative analysis [105], and experimental demonstration [99], to evaluate the performance of their system against the conventional solar absorber system and conventional RC system. All these studies used a PE film as the wind cover instead of a glass layer.

In another study, Hu et al. [106] investigated black acrylic paint as the a/e plate surface. In contrast to previous studies that had designed the plate to have a high emissivity only in AW (i.e., narrow emitter), the black paint used in this study had high emissivity in the entire MIR band (i.e., broadband cooler), as shown in Fig. 11d. A day-long experiment was conducted and it was found that the system collected heating energy of 8.64 MJ and cooling energy of 0.99 MJ.

Zhao et al. [107] proposed a polymer film (PDMS) as the RCer on top of a multilayered solar absorber. They conducted detailed optical simulations to identify the optimum thickness of materials to be used in the multilayer stack, in an attempt to reach the target value for the emissivity of an ideal a/e plate. The proposed surface had a strong solar absorption of 0.92 and emittance of 0.84 in AW. Nwaji et al. [108,109] first conducted a review and then simulated the performance of such a system in different Nigerian cities.

All aforementioned studies had considered a daytime solar heating and nighttime RC system in which the a/e plate had high absorptivity in the solar band and high emissivity in AW, close to the ideal profile (Fig. 11a). However, the high surface emissivity reduces the daytime thermal efficiency by around 15%−20%, as the absorbed heat is emitted out through thermal radiations [27]. This challenge can be addressed by having variable emissivity techniques, which are discussed below.

3.4.1. Self-adaptive module

The a/e plate of a daytime solar heating and nighttime RC system requires low emissivity in the MIR band (3−25 μm) for daytime solar heating and high emissivity in AW (8−13 μm) for nighttime RC. Previous studies had considered materials that lacked the ability of self-adaptive change based on the working time. To design such a plate that had variable optical properties, Ao et al. [110] investigated a self-adaptive plate using a thermochromic material (i.e., vanadium dioxide), which changes its optical properties in response to temperature variation as it undergoes phase transition [111]. Fig. 11e shows the absorptivity/emissivity profile for an ideal self-adaptive plate during the daytime when its temperature is above the critical temperature (T_c) of the material. During the night, when its temperature drops below T_c, the emissivity in AW increases, which is suitable for RC. Vanadium dioxide is a phase change material that exhibits intense changes in the optical properties near its critical temperature, which is a good feature to be used for self-adaptive thermal emission. Initial studies on using it for RC were made by Ono et al. [112] and Kort-Kamp et al. [113].

3.4.2. Double wind cover module

In contrast to previous studies, which considered a single unit a/e plate, Hu et al. [114] investigated a double-covered module, in which the absorber and emitter plates were separated from each other as shown in Fig. 11f. Ultra-white glass was used as the emitter because of its high emissivity in AW. The glass is highly transmissive in the solar band, and therefore allows solar radiation to be absorbed by the absorber plate underneath. As the absorber and emitter were arranged separately and not connected on a single plate, the high emissivity of the emitter (i.e., glass) does not reduce the thermal efficiency of the absorber plate. The module was designed and experimentally tested against a conventional solar absorber. Results showed that the highest temperature difference between the absorber plate and T_{amb} was 138.8 °C for the proposed module, whereas 134.2 °C for the solar absorber module.

3.4.3. Movable wind cover module

Vall et al. [115] developed a numerical model for a module consisting of a movable wind cover that could switch between two different materials (i.e., glass and PE) depending on transmission requirements of the working time. During the daytime, the glass cover was put on, as it had high transmittance in the solar band and negligible transmittance in other bands, which supports solar heating. For nighttime, the PE cover was put on, as it has high transmittance in all wavelength bands and favors heat emission from the emitter plate. Furthermore, a sensitivity analysis was conducted to identify the parameters that had a significant influence on system performance. It should be noted that the mechanism for switching the cover was not presented in this study.

3.4.4. Temperature regulating module

Liu et al. [116] proposed a temperature regulating module and studied its cooling performance for summers and heating performance for winters. The proposed module, as shown in Fig. 11g, consists of a PE film (wind cover), a porous cooling material (emitter), aluminum conductor, and chromium-plated solar absorber. The porous material had high emissivity in AW and could also scatter the sunlight due to the hierarchically porous structure. During the cooling mode for summers, the solar radiation got reflected by the porous structure, while the building's indoor heat got transferred to the module through conduction and convection (yellow) at the bottom side and was radiated to the sky through IR radiation (red). For the heating mode in winters, the module was turned upside down and solar radiation got absorbed by the solar absorbing materials and transferred to the building's indoor through IR radiation (red) as well as conduction and convection (yellow). Other than the aforementioned ways, many other techniques to passively and actively control the emissivity of RCers not specific to SE–RC systems have been reviewed in detail by Ulpiani et al. [117].

3.5. Daytime solar heating and radiative cooling

This section discusses the studies that investigated simultaneous solar heating and RC during the daytime. Chen et al. [118] first studied such a system using a germanium wafer solar absorber, which was highly transparent in the thermal wavelengths and was placed above an RCer (Fig. 12a). As solar shading was provided by the absorber plate on top, the RCer required few spectral constraints. To minimize convective losses, the RCer was enclosed inside a vacuum chamber that had a ZnSe window that is transparent in thermal wavelengths but is also very expensive and therefore difficult to be used for large-scale RC applications. Experimental results showed that the absorber plate reached a maximum of 24 °C above T_{amb}, while the cooler reached a minimum of 29 °C below T_{amb} (Fig. 12a). Instead of using SE for heating applications, the same system can be transformed to be used for electricity generation by replacing the solar absorber with PV cells that are transparent in thermal wavelengths.

Li et al. [119] designed and fabricated a dual-mode device using a film rolling technique, that could switch between the heating and cooling modes depending on the weather changes (Fig. 12b). The device had two materials, one with high absorptivity in the solar band for solar heating, and the other with high reflectivity in the solar band and high emissivity in AW for RC. Zinc film deposited with copper nanoparticles was used as heating material. PDMS coated on a silver film was used for RC, because of the high solar reflection of silver and high MIR emissivity of PDMS. Depending on the changes in weather and the building's internal demand, the required material (heating/cooling) would get exposed to the sky, while the remaining part would be rolled up and collected. Outdoor field tests showed a heating power and cooling power density of 643.4 W/m^2 and 71.6 W/m^2, respectively. Besides, a building energy simulation for predicting the performance in 16 different climatic zones of the USA showed that the device could save up to 236 GJ (19.2%) of energy annually for heating and cooling. Simultaneous solar heating and RC systems can have other design configurations and applications in the buildings sector, and therefore should be looked into for more future research.

4. Challenges and prospects

In this section, the current challenges faced by SE–RC systems, solutions investigated to resolve them, and recommendations for future research in this domain have been discussed.

4.1. Low-efficiency enhancement and energy output

Under realistic atmospheric conditions, replacement of conventional glass with ideal RCers does not have considerable PV temperature reduction. Therefore, instead of sole RC, many studies have combined SRC (section 3.1.2) and improved solar transmission (section 3.1.3) techniques to enhance the PV temperature reduction and improve electrical efficiency. Considering this, future investigation on RC of multijunction (tandem) solar cells could be a novel and interesting research area, as it would eliminate the stringent spectral requirements for RCers (i.e., sub-bandgap reflection).

As the nighttime cooling gain from SE–RC systems is not significantly high, different methods have been considered to address this challenge. These include the identification of optimum system design parameters [48,68] and atmospheric conditions [56,104]. Different micro/nanopatterned structures (e.g., pyramid [51], ant hair [67], micrograting [70]) have also been explored for RC surfaces. Ways to improve the heat transfer and finding the share of RC as an additional cooling resource to conventional systems have been investigated [73]. For DSH-NRC systems, variable spectral emissivity is a challenge and has been solved by considering different module configurations (section 3.4). New ways to improve the energy transfer from the a/e plate to the working fluid could be considered. Changing the heat transfer fluid (e.g., nanofluids) or the heat exchange surface (e.g., fins, surface roughness, flow configuration) are untapped research areas that can be investigated.

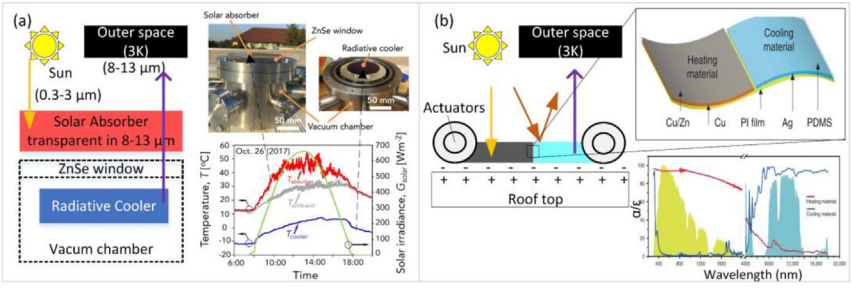


Fig. 12. **(a)** Schematic of the daytime solar heating and RC system considered in the study (on left). The solar absorber (red) is highly absorbing in the solar band (0.3–3 μm) and highly transparent in AW (8–13 μm). The ZnSe window is transparent in AW and is used to mechanically support the high vacuum (dashed black lines) that suppress the parasitic heat losses of the RCer. The RCer (blue) is highly emissive in AW. Schematic of the experimental setup (on the top right). The measured temperature of the solar absorber plate (red line), the RCer (blue line), and T_{amb} (grey line) during the experimentation (on the bottom right). Reproduced with permission and redrawn from Ref. [118]. **(b)** Schematic of the daytime solar heating and radiative cooling module considered in the study (on left). Structure of the dual-mode heating/cooling material (on the top right). Absorptivity/emissivity profile of the heating and cooling material (on the bottom right). Reproduced with permission and redrawn from Ref. [119].

4.2. Cost-cutting methods and economic assessment

The stringent requirements for spectral radiative properties make the fabrication process of RCers very expensive. In addition to finding cheap fabrication techniques, innovative installation methods should be considered that can help reduce overall costs. One way would be to integrate SE−RC systems into the building's skin to reduce additional material costs during building construction. This is analogous to the widely used building-integrated PVs [91], which are dual-purpose systems serving both as an outer layer of the structure and also generating electricity. Furthermore, research can be conducted in ways to spectrally modify the PV modules during their manufacturing time to eliminate the need for additional RCers. In addition to some recent studies [59,60,116] that have investigated the energy savings potential of SE−RC systems, a techno-economic analysis that considers costs, benefits, risks, and the time frame of such systems could also be considered.

4.3. Operating conditions and performance indicators

There are no uniform testing conditions and parameters to compare the performance of different types of RCers that have been used in the literature. Owing to the varying input conditions that are dependent on the geographic location, different numerical values have been used for input boundary conditions (i.e., G_{sun}, T_{amb}, and h_{conv}). Also, the PV temperature coefficient and maximum power conversion efficiency of the solar cells vary for each study. As for the cooling gain from the RCer, most studies have evaluated the net cooling power ($P_{net,rad}$), whereas others have presented the temperature difference between T_{amb} and RC surface (T_s) or a/e plate (T_{plate}). Also, as electrical and thermal energies have different thermodynamic qualities, the total system efficiency should be calculated using the more standardized exergy analysis instead of the conventional energy analysis [87]. Analogous to the standard test conditions for PVs ($T_{pv} = 25\ ^{\circ}C$, $G_{sun} = 1000\ W/m^2$, AM 1.5), future researchers should look into the development of standard conditions and performance indicators for RCers to better quantify their performance.

4.4. Field testing to assess the impact of climate and built environment

Most studies have carried out numerical simulations, whereas only a few have conducted outdoor field tests for just short periods (maximum 2−3 days). Season-long or year-long field testing could be considered as it would help evaluate the reliability and lifetime of SE−RC systems under realistic outdoor conditions. Also, changes in surface radiative properties from environmental influences such as dust accumulation or overnight dew formation could be investigated.

Researchers may investigate the performance of SE−RC systems in regions that lie near the equator, as they undergo very hot and long summers suitable for harvesting SE, and also for using RC gain because of the high cooling demands. Currently, studies have investigated the resource potential of individual SE and RC systems for some countries (i.e., USA [120], China [59], etc.). Also, regions that fall under the desert climate can be investigated because of their dry atmosphere, high solar irradiance, and low T_{amb} at nighttime, all of which support SE−RC systems.

All previous studies investigating RC have assumed an unobstructed view of the sky. However, in crowded urban centers, the presence of tall buildings obstructs the view and reduces the spatial window for heat loss to the sky [121]. The negative impact of radiative heat gain from the built environment has not been considered, and thus researchers can develop relevant models and experimentally verify this impact. Furthermore, in contrast to the extensively studied roof-mounted systems, investigating the performance of vertically mounted SE−RC systems can be an interesting research area.

4.5. Energy storage for variation in demand and supply

When using SE−RC systems, the variation in seasons and operating time creates a gap between the energy supply and demand. During summers, in the daytime, the cooling load demand is high but the RC gain is low. Excess cooling that may be achieved during nighttime can be stored and used the following day. During winters, at night, the heating demand is high but there is no solar heat gain. Excess heat energy available during the day could be stored for nighttime usage. Also, the RC gain harnessed at night is of little use. Considering these issues, the study of techniques to store excess heating and cooling energy should be investigated. Previously, phase-change materials have been integrated with a nighttime RC system for cooling energy storage [122], and in the future, other techniques such as thermochemical storage [123] can be investigated.

5. Conclusions

SE harvesting and radiative sky cooling are the relevant solutions to the ever-growing challenges of energy shortage and global warming. Over the years, there have been many developments made in these systems, to improve their performance, reliability, and longevity. However, as individual SE and RC systems face issues of limited working time and low energy gain, combined SE−RC systems resolve such issues and are a current research hotspot and promising future energy technology. In this review, we have thoroughly analyzed and discussed the studies that have investigated such systems from different perspectives. The different SE−RC systems studied so far are categorized into five major types based on their functionality and working time. Each type has been examined based on the system design, module configuration, operating conditions, and performance results. The similarities and differences between the studies were discussed, with a particular focus on the spectral radiative properties and materials used in each system. Details of materials, operating conditions, parameters investigated, and performance results are also provided in a tabulated form in the supplementary material.

RC has been extensively studied for non-concentrated PV systems; however, the few studies that investigated concentrated PV systems observed larger PV temperature drops. Furthermore, combining the selective reflection and improved transmission techniques to the RC of solar cells is found to be significantly useful in contrast to only RC. The impact of RC in other systems such as TPV system, PV-thermoelectric system, and CSP system have also been examined. Regarding the substitution of glass cover with highly emissive RC materials, differing viewpoints exist in the literature and so have been discussed by explaining the assumptions considered in the studies. To meet the spectral requirements of high thermal emittance in radiative coolers, solar absorptance in thermal collectors, and sub-bandgap reflection in PV cells, different types of materials including nanoparticle-based polymers, multilayered film stacks, and micro/nanopatterned structures have been investigated. For daytime solar heating and nighttime RC, dynamic radiators that change their spectral properties are most suitable and have been developed using different techniques like thermochromic materials and relevant wind cover designs. SE−RC systems have great potential and promising applications, but the major challenges to large-scale development include low cooling power and heat transfer, non-uniform operating conditions, and variation

in supply and demand. Possible solutions to these challenges and other corresponding issues have been proposed along with some recommendations for future researchers.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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